# MALLA REDDY UNIVERSITY

**B.Sc. Computer Science - III Year I-Sem** **L /T/P/C**

**4/0/2/4**

## (MR24-1BSCC17) STACK DEVELOPMENT FRAMEWORKS

## COURSE OBJECTIVES:

1. Understand the fundamental concepts and frameworks of Full Stack Development, including front-end, back-end, and database tools.
2. Analyse the architecture and key concepts of MongoDB and NoSQL
3. Explore the Django framework, its architecture, environment, and how to create basic web applications with views and templates.
4. Implement forms, user input handling, and user authentication in Django, including CRUD operations and session management.
5. Develop RESTful APIs using Django Rest Framework, understanding REST architecture and creating API endpoints for data manipulation.

## CONTENT

### UNIT- I

**Introduction to Full Stack Development Frameworks:**

Full Stack Development, Need of Full Stack Development, Introduction to Front end Environment Tools, Introduction to Backend Environment Tools, Introduction to Databases, Client/Server architecture structure.

**Stack Frameworks:**

MEAN, MERN, MEVN, LAMP, Rails or Ruby on Rails, Django Stack

### UNIT-II

**MongoDB**

Understanding NoSQL and MongoDB, Getting Started with MongoDB and Node.js, Manipulating MongoDB Documents from Node.js, Accessing MongoDB from Node.js, Using Mongoose for Structured Schema and Validation, Advanced MongoDB Concepts.

## UNIT-III

**Django Frameworks**:

Django Architecture, Features, Web Server, Understanding Django environment, Installing & Configuring Django Components, Hello World Application

**Views and Templates**

Creating views and mapping to URLs, introduction to template engines and creating templates, rendering dynamic content in templates, inheritance for DRY templates

## UNIT–IV

**Forms and User Input**

Building forms using Django's, Form API, Handling user input and validation, implementing form submission and processing, form widgets and handling form errors

**Authentication and User Management**

CRUD operations, implementing user registration and authentication, managing user sessions and permissions, customizing the authentication system, Building user profiles and account management

## UNIT-V

**RESTful APIs with Django Rest Framework:**

Introduction to REST architecture, Building Web API using Django REST, setting up Django Rest Framework (DRF), Creating API endpoints for data retrieval and manipulation.

## TEXT BOOK:

1. Django Unleashed by Andrew Pinkham, Copyright © 2016 by Pearson Education, Inc.
2. MongoDB: The Definitive Guide by Kristina Chodorow and Michael Dirolf.
3. NoSQL Distilled by Pramod J. Sadalage and Martin Fowler.
4. Full Stack Web Development: Round One – Begin!By Edwin Ross Torres

## REFERENCE BOOKS:

1. The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Stack Web Developer by Chris Northwood
2. MongoDB Basics by Peter Membrey and Tim Hawkins.

3. Full Stack Django and React: Get hands-on experience in full-stack web development with Python, React, and AWS by Kolawole Mangabo

## COURSE OUTCOMES:

1. Identify and describe Full Stack Development frameworks and their components, including MEAN, MERN, MEVN, LAMP, Rails, and Django.
2. Demonstrate proficiency in MongoDB and NoSQL.
3. Create basic web applications using Django, including setting up the environment, creating views, and rendering templates.
4. Develop forms and handle user input in Django, implementing user authentication, session management, and CRUD operations.
5. Build RESTful APIs with Django Rest Framework, setting up DRF, and creating API endpoints for data retrieval and manipulation.

## STACK DEVELOPMENT FRAMEWORKS LAB

## LIST OF LAB PROGRAMS:

1. Demonstrate version control in Git and Github using simple html code
2. Create a simple Node.js server with routes login, register, profile and logout
3. Write MongoDB model for the user and query to fetch user, validate and register.
4. Write a server side program for Accessing MongoDB from Node.js.
5. Write a server side program for Manipulating MongoDB from Node.js.
6. Install Django and setup a virtual environment. Display Hello World Application
7. View a List of Books with Titles and Authors Using Django
8. Design Web Application with different menu items using Django.
9. Handle user input and perform validation, then implement form submission and processing in Python.
10. Fetch user details from server using REST API and show in profile menu using Django.